DARSHAN H

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Experience

Samsung R&D Institute India - Bangalore **Jul 2025 - Jan 2026**

Project Developer Intern, PRISM | Bengaluru, India

- Developed and maintained **React.js features** in an internal CRM; translated Figma designs into reusable UI components and integrated REST APIs.
- Collaborated with the Samsung team to implement features, troubleshoot issues and participate in code reviews before delivery.
- Applied modular component architecture and Git-based workflows to support maintainable code and collaborative software development.

Lagnam Matrimony - Client Application [Play Store] **Delivered in 5 days**

Freelance Application Developer | React Native/Expo, Node.js, MongoDB

- Built a client mobile application end to end in **five days** using AI-assisted development; deployed its Node.js/Express backend to a **public VPS** and completed the application setup.
- Integrated MongoDB-backed profiles, preference-based recommendations, authenticated Socket.IO chat and Razorpay payments with verification and duplicate-payment checks.

Projects

Glass Cage 2.0 | Node.js, TypeScript, Python, React, Three.js, Linux eBPF

Zero-trust runtime supply-chain sandbox and dependency firewall

- Built and deployed a runtime dependency firewall for Node.js and Python, combining asynchronous package attribution with Linux seccomp/eBPF to intercept sensitive-file access and unauthorized network calls.
- Implemented canary honeytokens, DNS-entropy alerts, and per-package access policies; integrated a live React/Three.js GeoIP dashboard with stack-trace inspection and connection approval controls.

CrashForge | Python, SQLite, pytest, CLI

Chaos testing and failure reduction engine for tool-using AI agents

- Built and released a Python CLI for chaos-testing tool-using AI agents, injecting after-commit timeouts, stale reads, and malformed results; detected duplicate side effects through independent ledger-based invariants.
- Implemented budgeted delta debugging to reduce fault plans while preserving failure signatures; persisted SQLite/NDJSON evidence and exported offline regression tests with replay-divergence checks for LLM traces.

AI Biometric Attendance System | Python, scikit-learn, dlib, PostgreSQL

Multimodal identity checks and role-based attendance management

- Built student identification using **128-dimensional facial embeddings and an SVM classifier**; combined face identification with speaker verification using librosa and Resemblyzer.
- Connected teacher/student workflows to biometric records and attendance logs in Supabase PostgreSQL; added bcrypt password hashing and QR-code class onboarding. **[Code]**

Technical Skills

Languages & Data: Python, JavaScript, TypeScript, SQL, NumPy, pandas
Machine Learning: scikit-learn, Regression, SVMs, Trees & Ensembles, Clustering, Feature Engineering, Model Evaluation
Deep Learning: PyTorch, Neural Networks, CNNs, RNNs, GANs, Transformers, Model Training & Evaluation
Generative AI & NLP: LLMs, Attention, Embeddings, RAG, Tool-Using Agents, Agent Evaluation
Full Stack (MERN): MongoDB, Express.js, React, Node.js, Next.js, REST APIs, WebSockets, React Native/Expo, Flask
Software Engineering & Tools: DSA, OOP, DBMS, OS, Networks, Git, pytest, Docker, AWS, Linux, PostgreSQL, SQLite

Leadership & Awards

- **First Prize, INR 25,000 - CITI-ZEN 2026 TERRABYTE Hackathon:** Co-developed Parisar-Netra, an AI/IoT campus energy-management prototype integrating ESP32 relay control, a MERN dashboard, TensorFlow.js occupancy detection and Gemini-assisted energy audits.
- **Club Lead - Algorand Blockchain Club CIT (Sep 2025 - Jun 2026):** Organized technical workshops and blockchain bootcamps with student contributors and faculty advisors.

Education

Cambridge Institute of Technology **2023 - 2027**

Bachelor of Engineering in Information Science & Engineering | Bengaluru, India